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1: SHEET PREPARATION AND TECHNICAL LETTERING

- 1) Prepare Drawing sheet
- 2) Write freehand, in single stroke vertical capital letters, small letters and numerals, using the ratio 7:5.

2: APPLIED GEOMETRY

2.1 Geometrical construction

- 1) Trisect 100mm long straight line.
- 2) Divide 100mm long straight line in to 6 equal parts.
- 3) Divide 100mm long straight line in to 2:3:4 ratio.
- 4) Trisect an acute angle.
- 5) Trisect an obtuse angle.
- 6) Construct a regular pentagon, inscribed in a circle of diameter 70mm.
- 7) Construct a regular pentagon having sides 40mm.

2.2 Tangents

- 1) Construct an external (uncrossed) common tangent between two different circles of diameters 60mm and 30mm respectively, where center to center distance of circles is 70mm.
- 2) Construct an internal (crossed) common tangent between two different circles of diameter 60 mm and 30mm respectively, where center to center distance of circles is 70.

2.3 Conic section

- 1) Construct an ellipse by using four center methods, where the major axis is 100mm and minor axis is 70 mm.
- 2) Draw a parabola by using triangular/tangent method where base of parabola is 70mm and height of axis is 60mm.
- 3) Construct a hyperbola when the distance of the focus from directrix is 35mm and eccentricity is $\frac{3}{2}$. (Definition /eccentricity method)

2.4 Curves and helix

- 1) Draw a cycloid of circle of diameter 40mm.
- 2) Draw an Archimedean's spiral of one convolution, where the radius of circle is 40mm.
- 3) Draw involutes of regular pentagon of sides 20mm.
- 4) Draw involutes of circle of diameter 30mm.
- 5) Draw a cylindrical helix of one convolution, where the diameter of cylinder is 40mm and height is 60mm.
- 6) Draw a conical helix of one convolutions, where the base diameter of cone is 50mm and height is 80mm.

3: BASIC DESCRIPTIVE GEOMETRY

3.1 Projection of Points

Draw the projections of the following points

- 1) Point A, 30mm in front of VP and 35mm above HP
- 2) Point C, 20mm behind VP and 25mm above HP.
- 3) Point D, 35mm behind VP and in the HP.
- 4) Point E, 20mm behind VP and 30mm below HP
- 5) Point F, in the VP and 40mm below HP

3.2 Projection of Lines.

Draw the projection of the following lines.

- 1) Line AB, 40mm long, parallel to HP and VP both, when the distance from HP and VP is 20mm and 30mm respectively.
- 2) Line CD, 45mm long contained by HP, and perpendicular to VP, when one of its end is 20mm away from the VP.
- 3) Line EF, 50mm long, contained by HP and inclined to VP at 45° , when one of its end is 15mm from the VP.
- 4) Line KL, 40mm long, contained by both HP and VP.
- 5) Draw the projection of the line MN when its end M is 10mm from the HP and 15mm from the VP and end N is 30mm from the HP and 40mm from the VP. Its end projectors are 40mm apart.
- 6) The front view of line inclined at 30° to the VP is 60mm long. Draw the projection of the line, when it is parallel to and 40mm above the HP, its one end being 30mm in front of the VP.

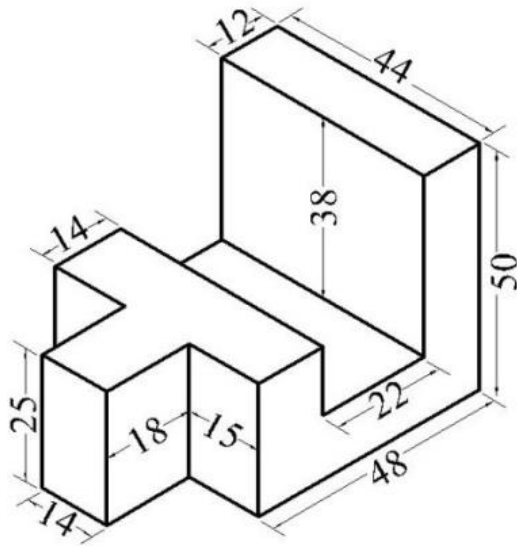
3.3 Projection of Planes.

Draw the projection of the following Planes.

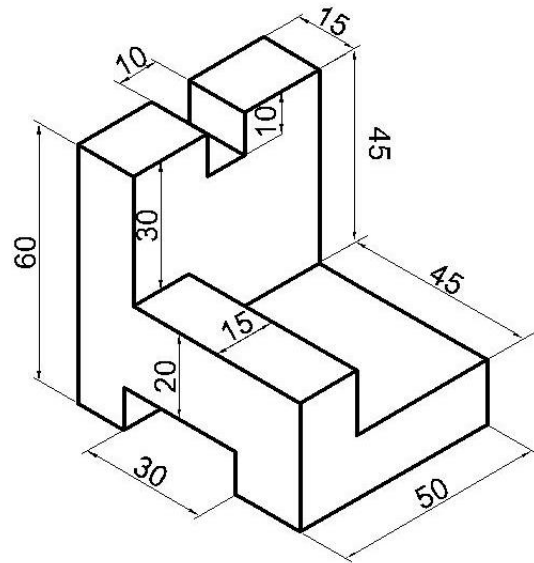
- 1) A square lamina ABCD, having 25mm side is parallel to HP and is 10mm from it. Its side nearer to VP is parallel to and 10mm from VP. Draw its projection.
- 2) A square plane ABCD of 40mm side is parallel to HP with one of its side inclined at 60° to VP and one side of square close to the VP is 15mm in front of VP and 20mm above the HP. Draw its projections.
- 3) A regular pentagon ABCDE, of 20mm sides has its one corner A in the HP and the side CD is parallel to HP. Draw its projections when its plane is parallel to and 10mm from the VP.
- 4) A regular hexagon of 25mm side has one side in HP. The plane is inclined at 30° to the VP and perpendicular to the HP. The nearest corner to the VP is 20mm in front of VP. Draw the projections of the plane..
- 5) Draw the projection of a thin circular sheet of 50mm diameter and negligible thickness, when its plane is inclined at 45° to the VP and is perpendicular to the HP. A point on its circumference and nearest to the VP is 40mm away from HP and 14mm from the VP.

3.4 Orthographic Projection

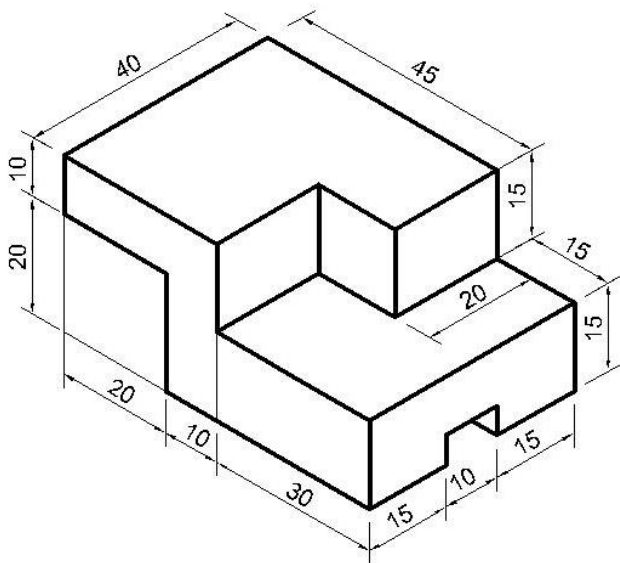
Draw complete orthographic views (Top, Side & Front) of the given solid figures.



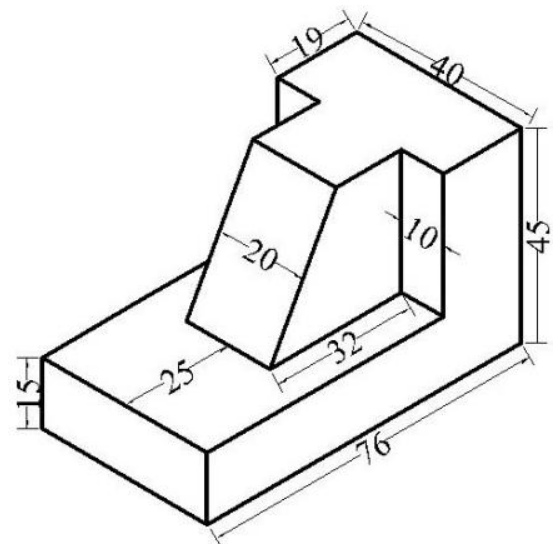
(Figure: 3.4 -1)



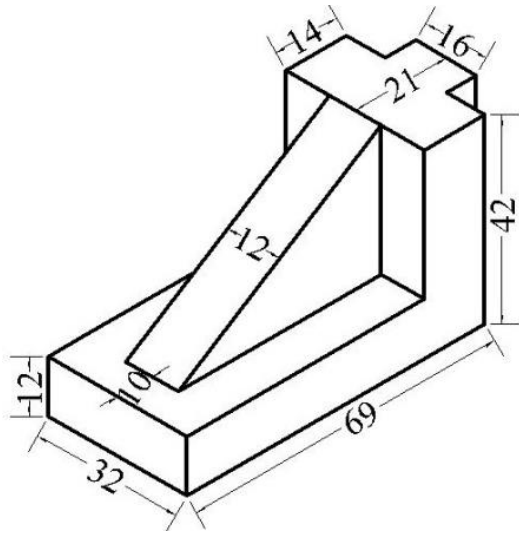
(Figure: 3.4 -2)



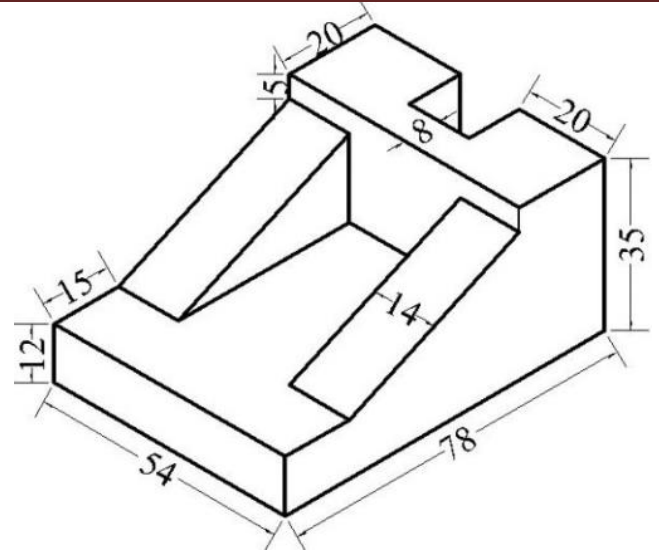
(Figure: 3.4 -3)



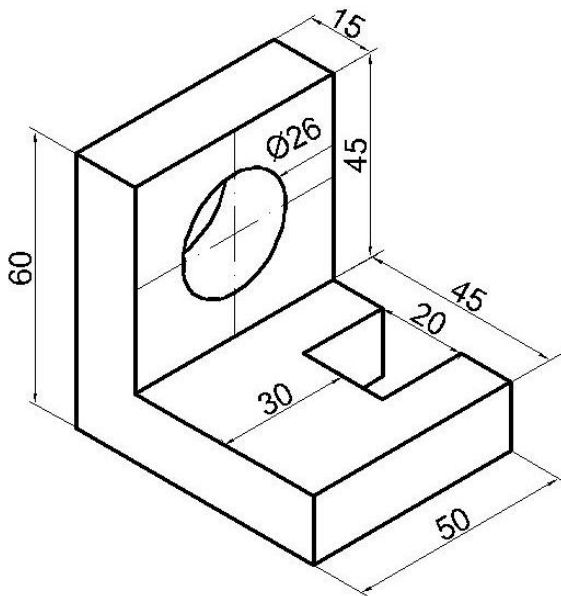
(Figure: 3.4-4)



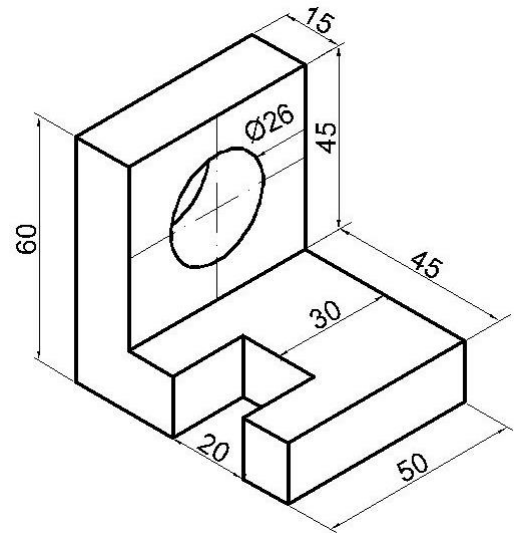
(Figure: 3.4-5)



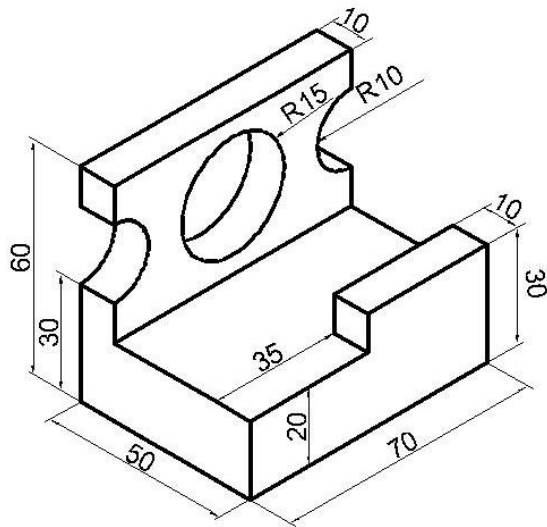
(Figure: 3.4-6)



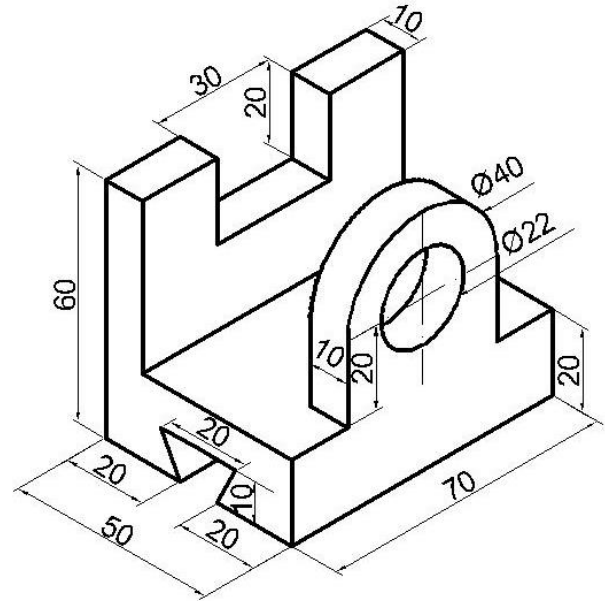
(Figure: 3.4-7)



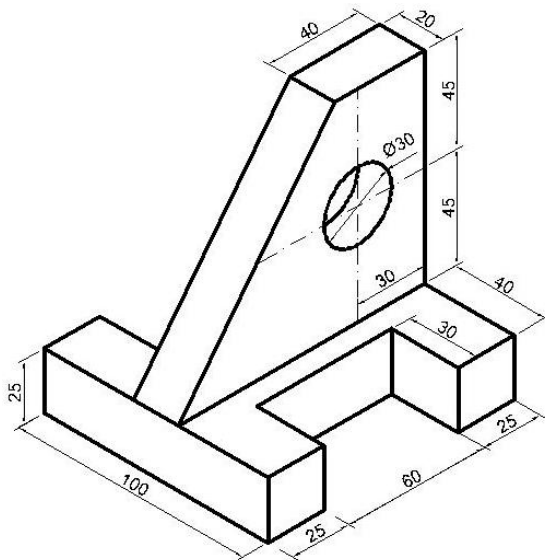
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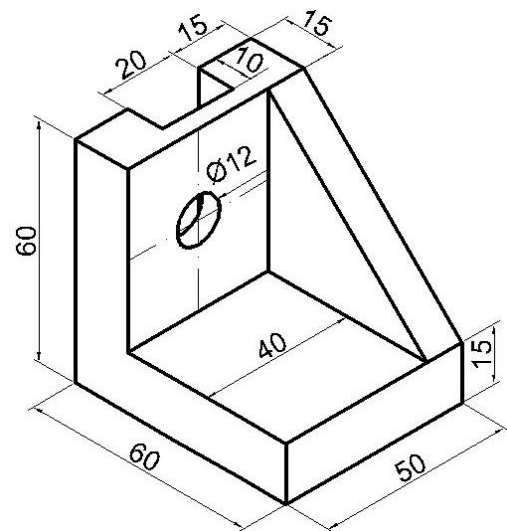
(Figure: 3.4-9)



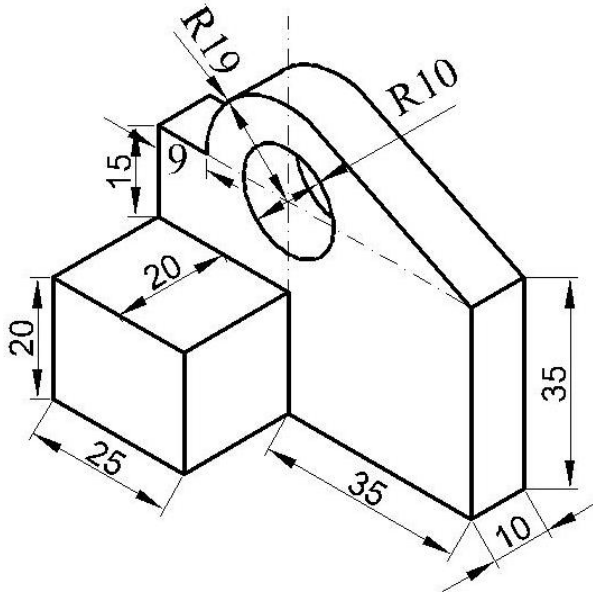
(Figure: 3.4-10)



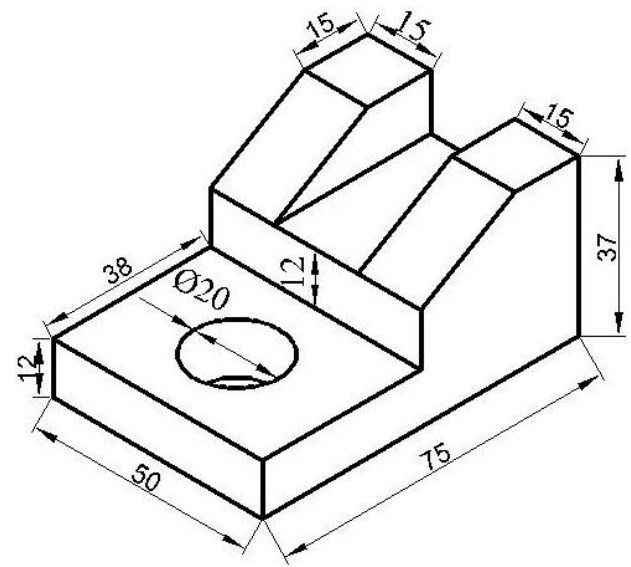
(Figure: 3.4-11)



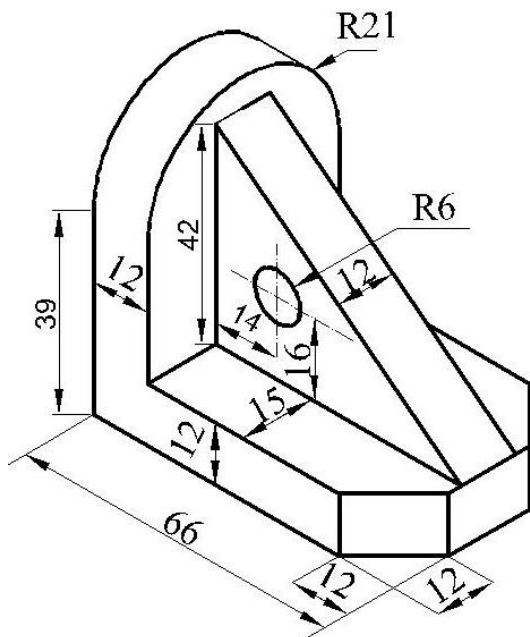
(Figure: 3.4-12)



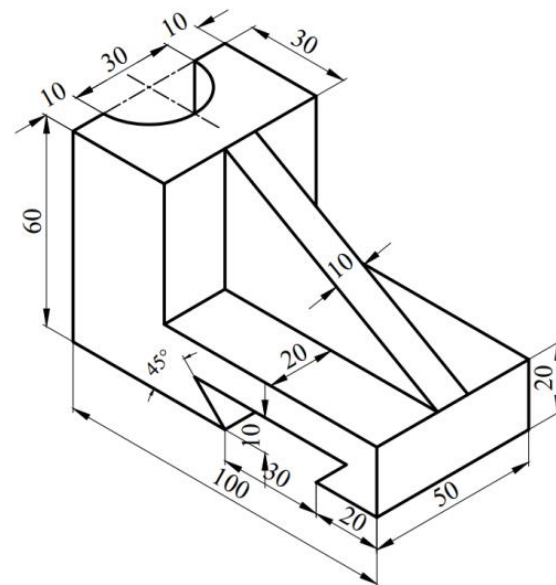
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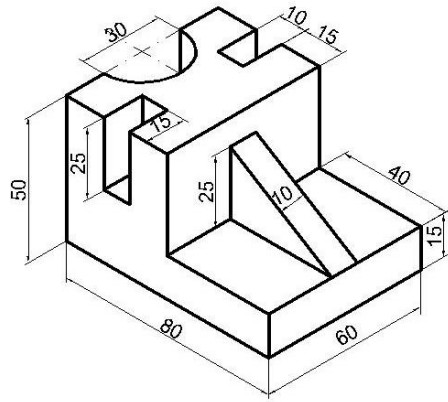
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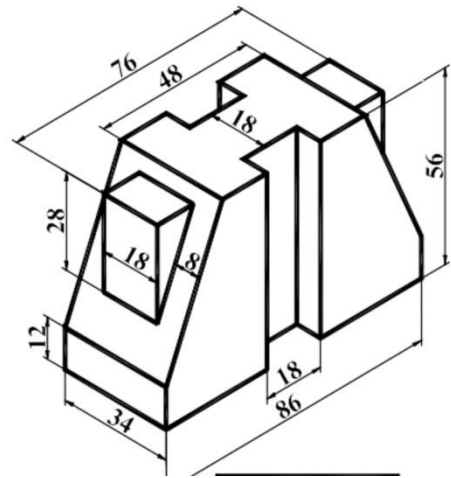
(Figure: 3.4-15)



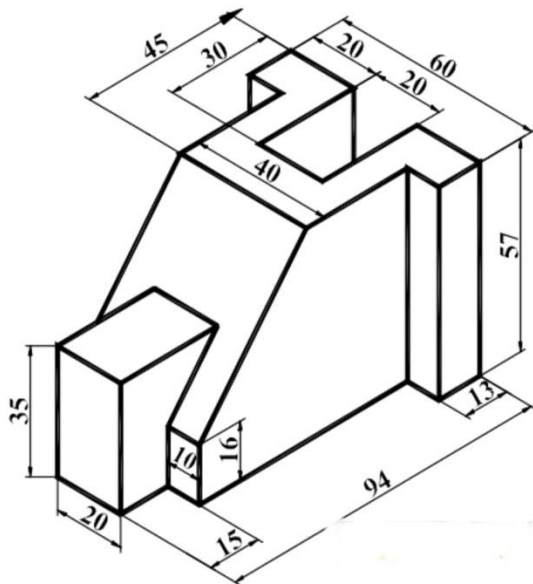
(Figure: 3.4-16)



(Figure: 3.4-17)

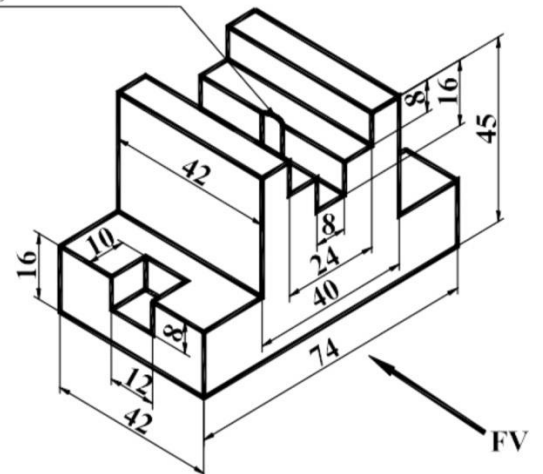


(Figure: 3.4-18)

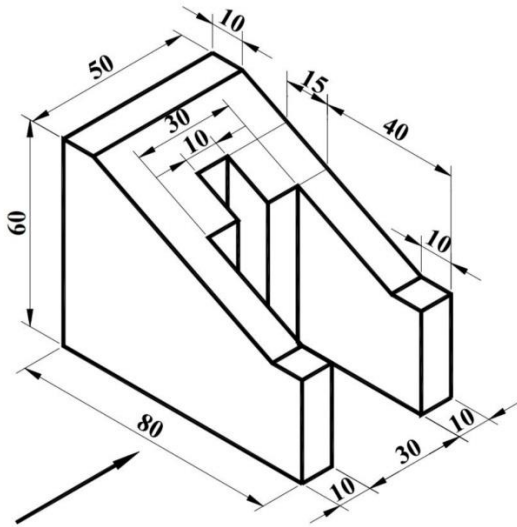


(Figure: 3.4-19)

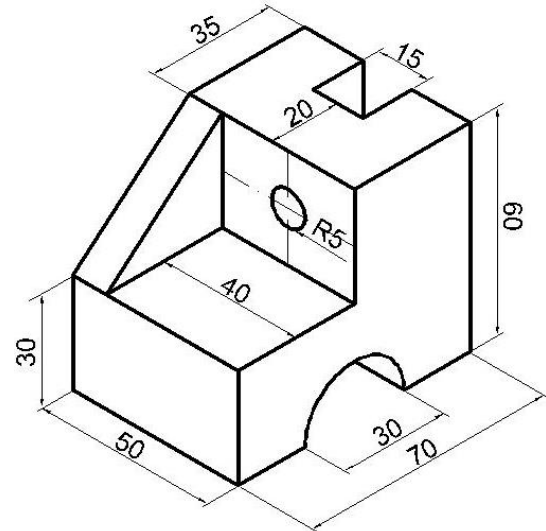
Through Hole 10 Dia.



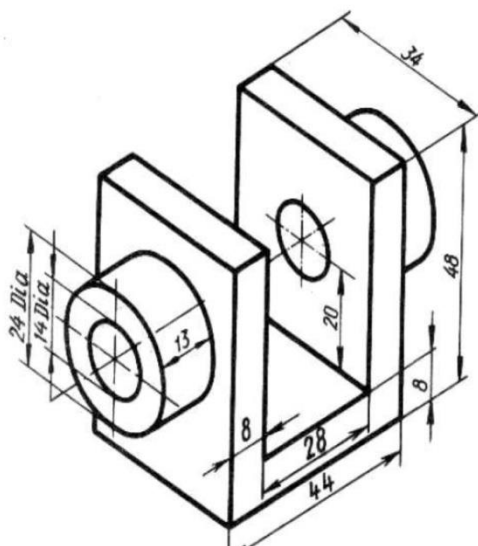
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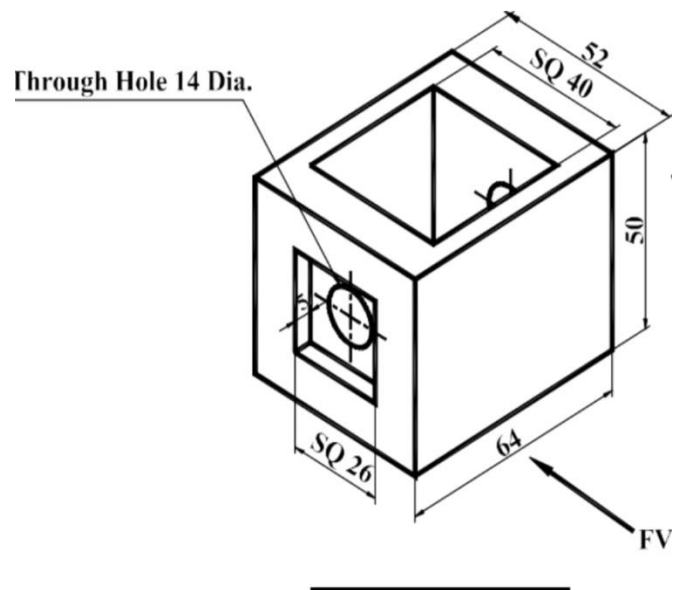
(Figure: 3.4-21)



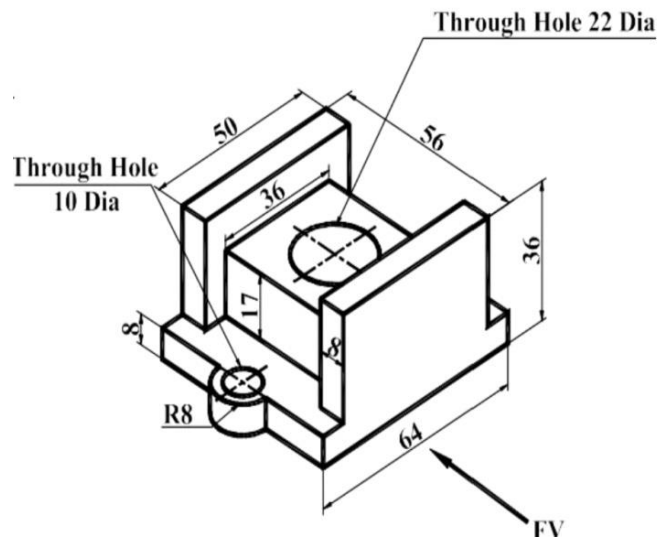
(Figure: 3.4-22)



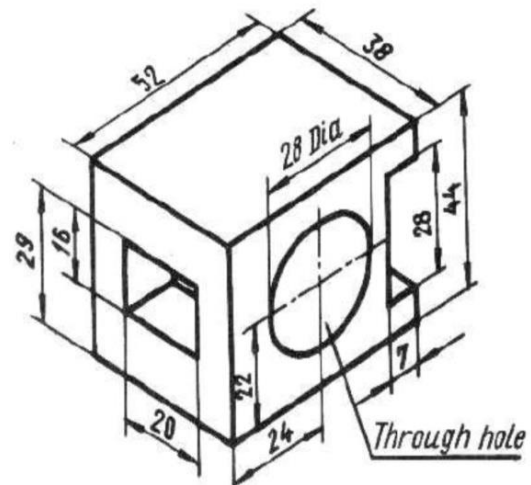
(Figure: 3.4-23)



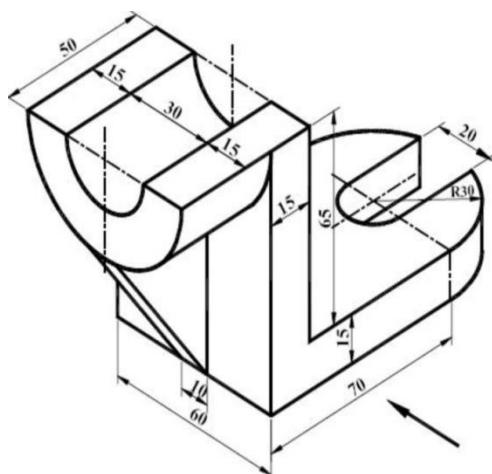
(Figure: 3.4-24)



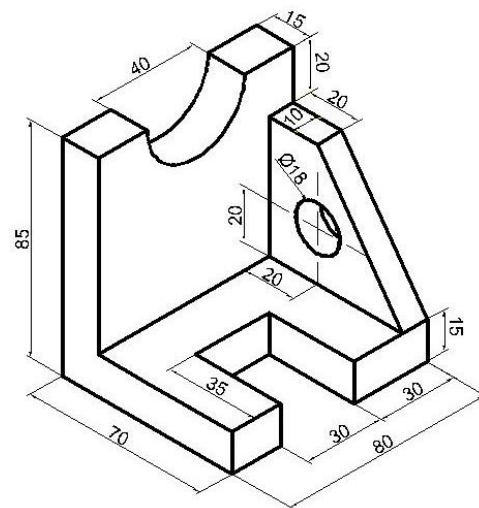
(Figure: 3.4-25)



(Figure: 3.4-26)



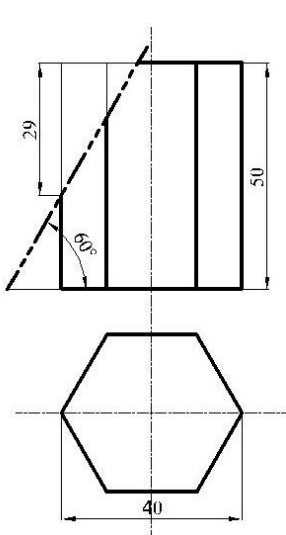
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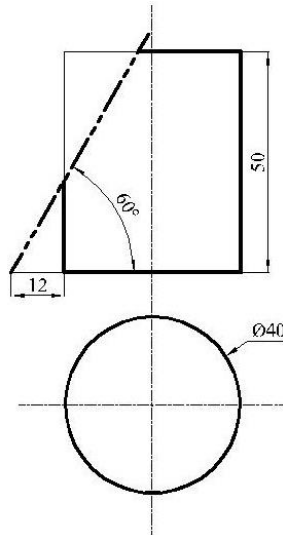
(Figure: 3.4-28)

4: SURFACE DEVELOPMENT

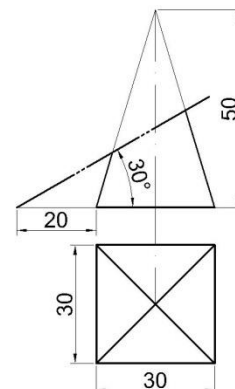
If a cutting plane cuts the given objects in to given position, as shown in figures. Reproduced the given views of the objects and draw complete surface development and also draw the true shape of the section.



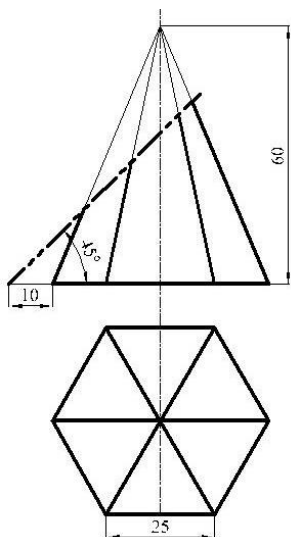
(Figure: 4-1)



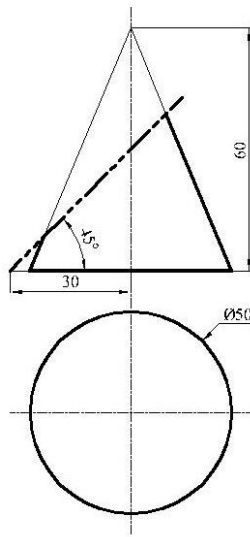
(Figure: 4-2)



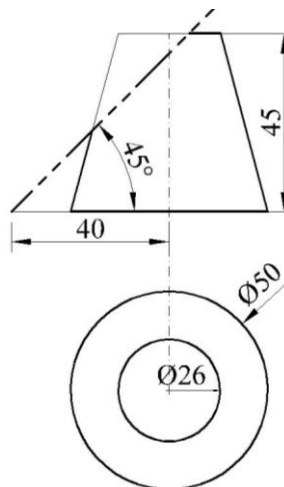
(Figure: 4-3)



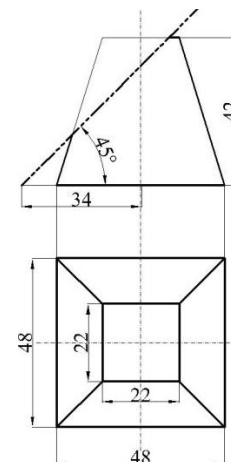
(Figure: 4-4)



(Figure: 4-5)



(Figure: 4-6)

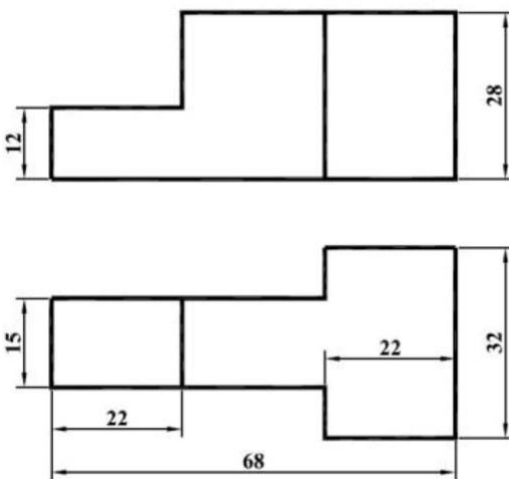


(Figure: 4-7)

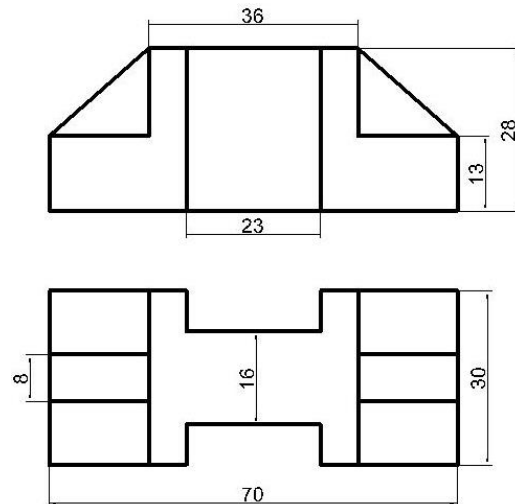
5: PICTORIAL PROJECTION

5.1 Isometric Projection

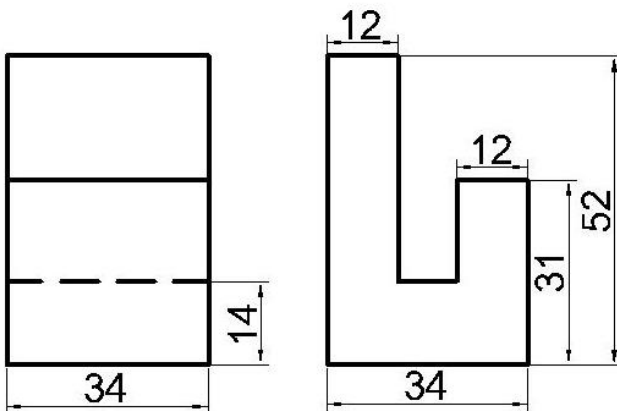
Draw an isometric view from the given orthographic views.



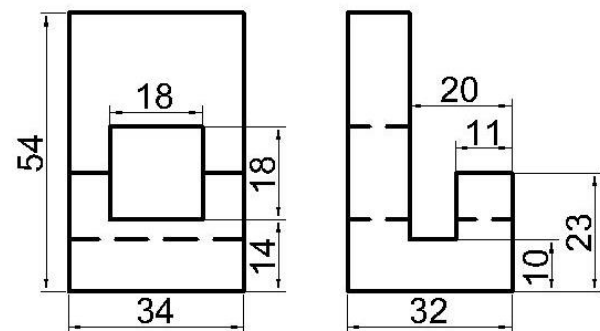
(Figure:5.1-1)



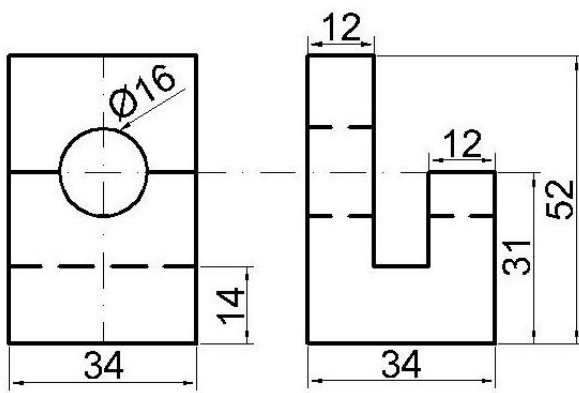
(Figure:5.1-2)



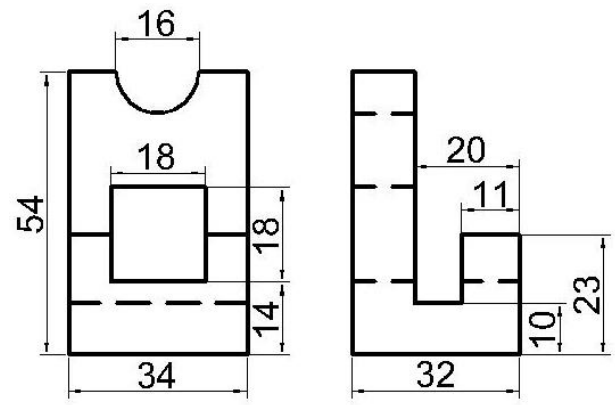
(Figure:5.1-3)



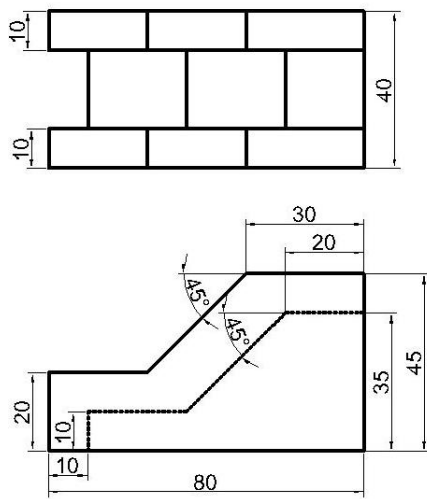
(Figure:5.1-4)



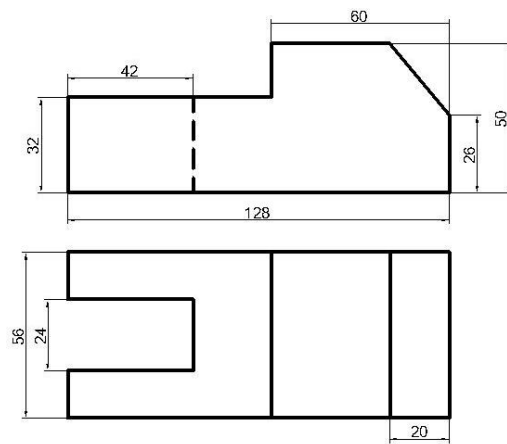
(Figure:5.1-5)



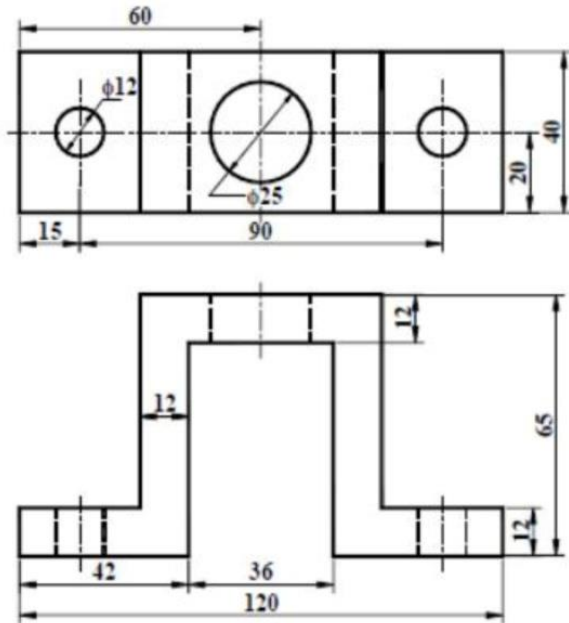
(Figure:5.1-6)



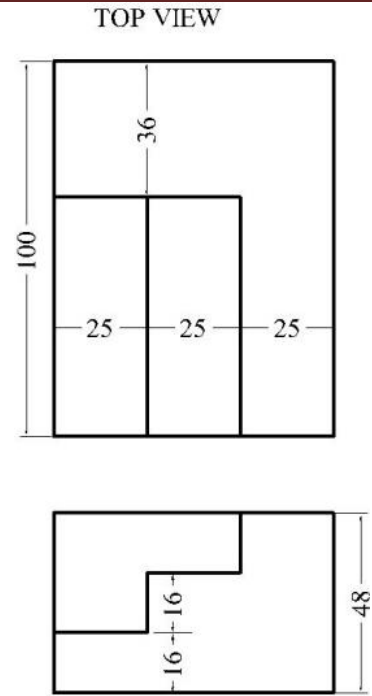
(Figure:5.1-7)



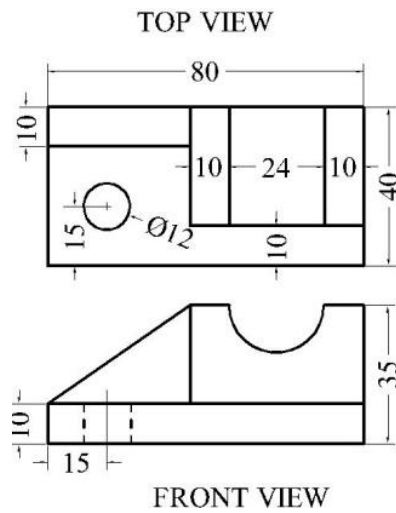
(Figure:5.1-8)



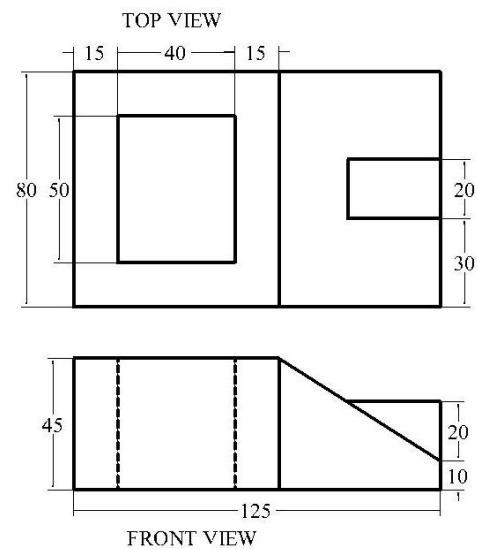
(Figure: 5.1-9)



FRONT VIEW
(Figure: 5.1-10)



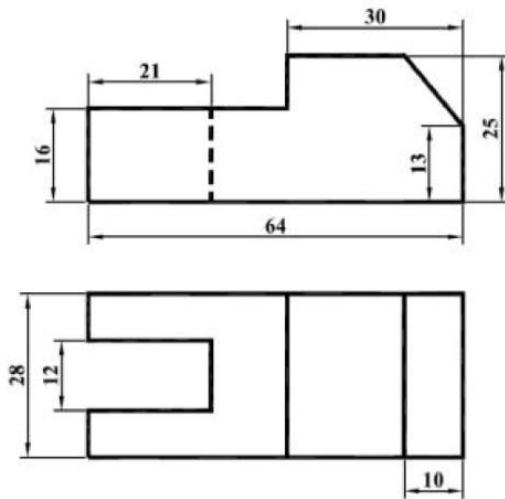
(Figure: 5.1-11)



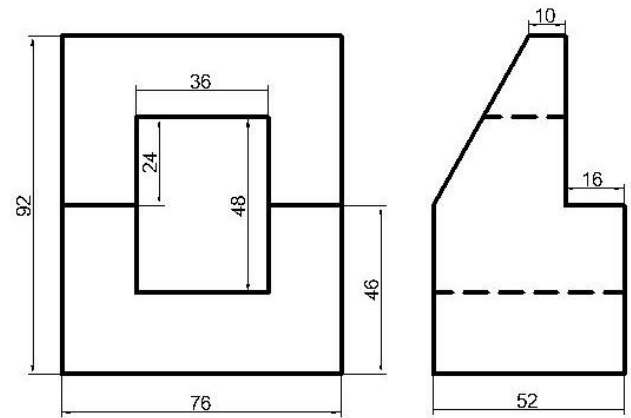
(Figure: 5.1-12)

5.2 Oblique projection

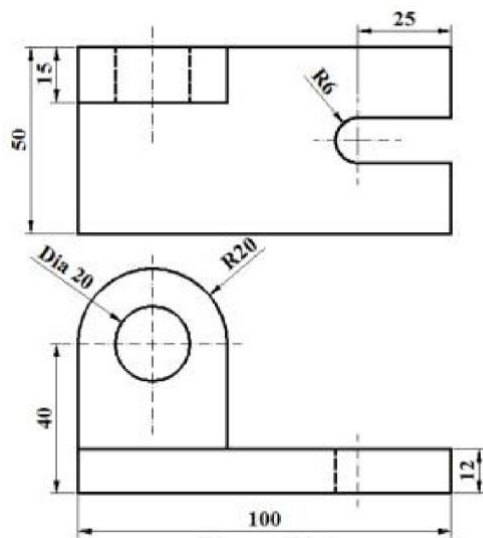
Draw an oblique view from the given orthographic views.



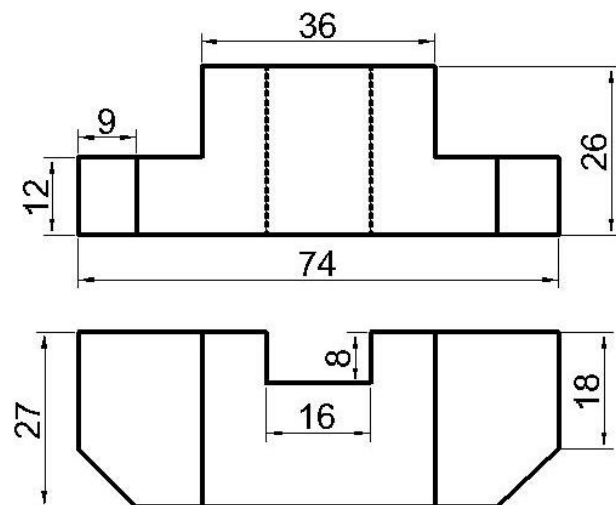
(Figure: 5.2-1)



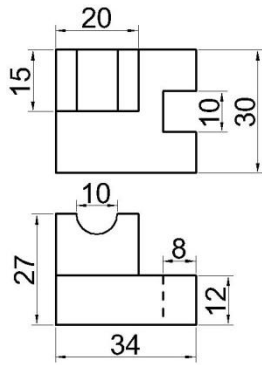
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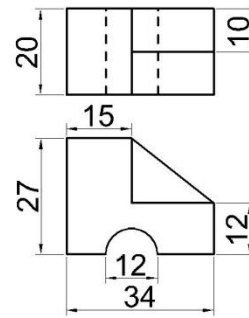
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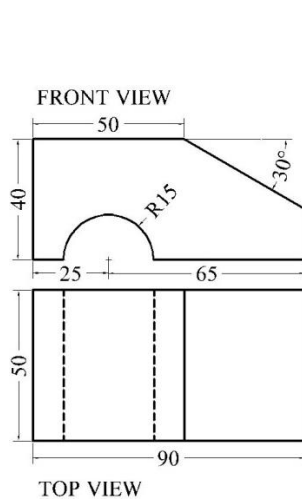
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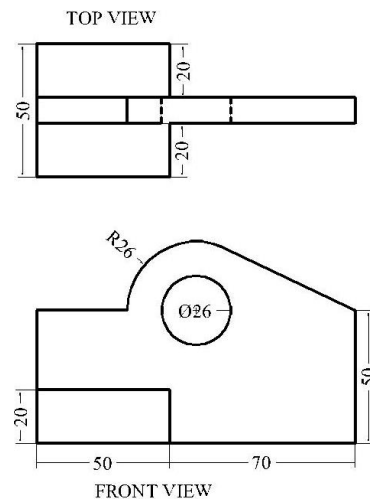
(Figure: 5.2-5)



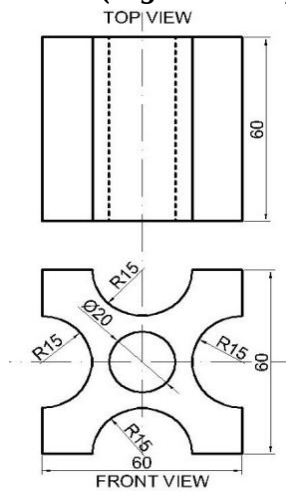
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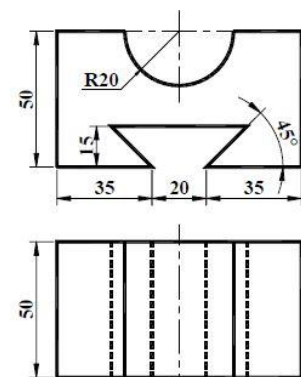
(Figure: 5.2-7)



(Figure: 5.2-8)



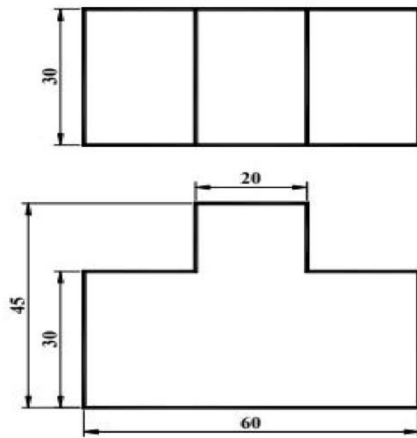
(Figure: 5.2-9)



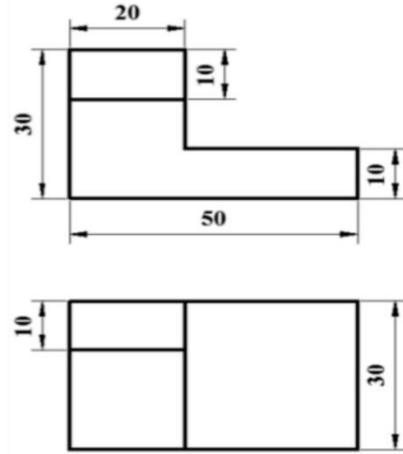
(Figure: 5.2-10)

5.3 Perspective Projection

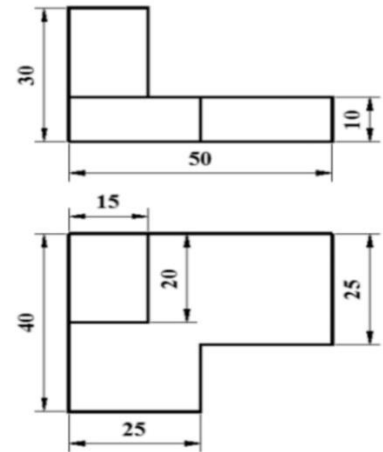
Draw the perspective projection from the given orthographic views.



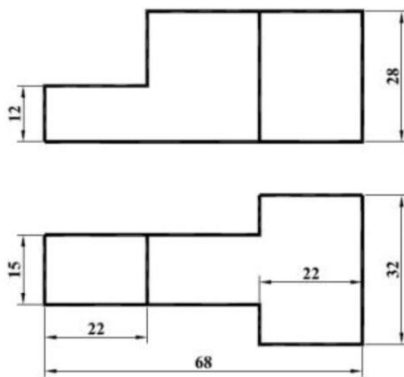
(Figure: 5.3-1)



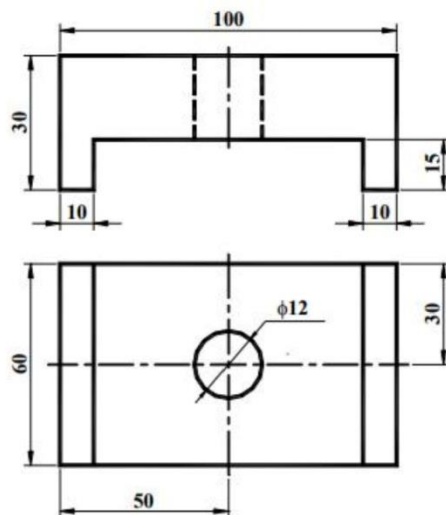
(Figure: 5.3-2)



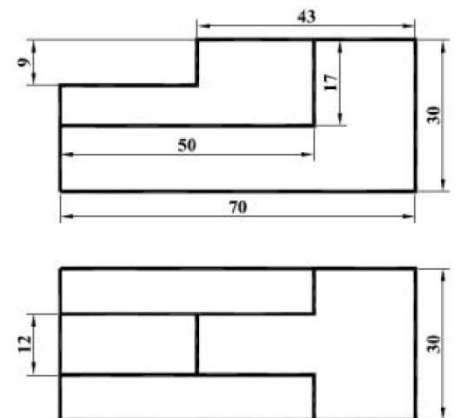
(Figure: 5.3-3)



(Figure: 5.3-4)



(Figure: 5.3-5)



(Figure: 5.3-6)

6:GRAPHICAL SYMBOLS

Sketch free hand the graphical symbols for the following Engineering items.

Electronics and Electrical

Amplifier		Antenna		Arrester		Battery	
Circuit Breaker		Capacitor		Coil		Connector	
Electric Contact		Core		Fuse		Directional Coupler	
Ground		Handset		Rectifier		Visual Signaling device	
Receiver		Repeater		Ground		Handset	
Thermocouple		Inductor Winding		Incandescent Lamp		Ballast Lamp	
Transformer		Switch		Resistor		Transmission Path	
Generator		Motor		Solenoid		Thermostat	
Thermistor		Variable Resistor		Voltmeter		NPN-type Transistor	
PNP-type Transistor		Direct Current		Alternating Current		Power Frequency	
Apparatus & Machine suitable for DC or AC		Half-wave Rectifier		Line or Cable existing		Line or Cable planned	
Controlled Rectifier		Power Line		Underground Cable		Overhead Line	

Electronics and Electrical

Conductors or a group of Conductors		Flexible Conductors		Two Conductors		Three Conductors	
Four Conductors		Junction of Conductors		Crossing without Electrical Connection		Crossing and Connecting Conductors	
Frame and Chassis Connection		Fault		Fault to Frame		Earth Fault	
Mechanically Coupled Machine		DC Generator		AC Generator		DC Motor	
AC Motor Single Phase		Three Phase Motor		Three Phase Motor in Delta Connection		One Way Switch (Single Pole Switch)	
Two Pole Switch		Three Pole Switch		Two Way Switch		Intermediate Switch	
Push Button Switch		Socket Outlets		Socket Outlets, 5A		Socket Outlets, 5A with Switch	
Socket Outlets, 15A with Switch		Lamp mounted on ceiling		Group of 3, 40 Watt Lamp		Lamp mounted on wall	
Fluorescent Lamp		Ceiling Fan		Wall mounted Fan		Exhaust Fan	
Fan Regulator		Bell		Pickup		Buzzer	
Siren		Public addressing System		Diode with Filament		Telephone- Telegraph Line	